

AI Trip Planner — How It Works

Step-by-Step Flow | 11 Agents | 3 Decision Gates | LangGraph + Flask

STEP 1 — User Fills the Trip Form

Enters: Source, Destination, Dates, Budget, No. of Travelers, Hotel & Transport preferences, Interests, Must-Visit Places.

STEP 1b — OR — Chats with Trippy (AI Chatbot)

User describes the trip in natural language. Trippy collects all details through conversation and auto-fills the form using GPT-4o-mini.

STEP 2 — Input Validation (Guardrails)

Checks: Destination filled? Budget $\geq$ Rs.500? Days $\geq$ 2? Source $\neq$ Destination? No harmful input? FAIL -> shows error instantly. PASS -> workflow starts.

[GATE 1] Are all required fields present in state?

YES -> Continue to Memory Agent | NO -> Stop. Show error to user.

STEP 3 — Memory Agent — Load User History

Checks if this user has planned trips before. If yes, loads saved preferences (e.g. "prefers seafood", "likes beach hotels") to personalize this trip.

STEP 4 — 4 Data Agents Collect Real Information

Weather Agent -> OpenWeatherMap -> daily forecast, temperature, rainfall risk Transport Agent -> Serper API -> flights/trains/buses, prices, departure hub Hotel Agent -> Serper API -> hotels matching preference and budget tier Places Agent -> Serper API -> attractions, restaurants, nearby day trips

STEP 5 — Budget Agent — Calculate and Auto-Optimize

Total = transport + hotel x nights x rooms + food x days x travelers + activities + 10% misc. Over budget? -> Tries cheaper hotel tier. Still over? -> Tries cheaper transport. Shows popup if still over.

[GATE 2] Any conflicts in the collected data?

NO conflicts -> Build the itinerary | Conflict found (rain+outdoor / hotel too pricey) -> Flag and continue

STEP 6 — Itinerary Agent — AI Writes the Day-by-Day Plan

GPT-4o-mini receives all real data and writes a complete time-blocked itinerary. Includes must-visit places. Schedules nearby day trips every 2 days. Validates output — rejects hallucinated wrong-city place names.

STEP 7 — Review Agent — AI Reads Plan as a Travel Expert

Llama 3.3 checks: Does plan match budget? Weather? User preferences? Returns: review paragraph + issues list + highlights list + verdict (Approved / Needs Retry).

STEP 8 — Evaluation Agent — AI Scores the Plan 0 to 100

Budget Fit (25 pts) + Completeness (25 pts) + Weather Alignment (20 pts) + Data Quality (15 pts) + Transport Logic (15 pts) Grade: A = 90+ B = 75+ C = 60+ D = below 60

[GATE 3] Is the generated plan good enough?

APPROVED -> Save to memory + Generate PDF | ISSUES FOUND -> Retry pipeline (max 3 times)

STEP 9 — Memory Agent — Save This Trip

Saves completed trip and preferences to memory/user_memory.json. Next time this user plans a trip, their preferences are remembered automatically.

STEP 10 — PDF Generator — Create Downloadable Trip Report

Generates full PDF using ReportLab: cover page, budget table, day-wise itinerary, hotel & transport details, weather summary, packing checklist, emergency contacts.

STEP 11 — Results Shown on the Web Dashboard

User sees: Hotels, Transport, Places, Day-wise Itinerary, Budget Breakdown, Score Card. Smart popups: Weather warning, Budget exceeded, No tickets available. Download PDF button.

SUMMARY

Quick Reference

Component	Technology	Job
Web Server	Flask	Receive form, run workflow, return JSON
Workflow Engine	LangGraph	Connect all agents, handle routing decisions
Decision Gates	Python if/else	3 checkpoints that stop or redirect the flow
Data Collection	Serper + OpenWeatherMap	4 agents fetch real live data in parallel
Cost Calculation	Rule-Based Math	Budget agent calculates and auto-optimizes
Plan Writing	GPT-4o-mini	Itinerary agent writes the day-by-day plan
Quality Check	Llama 3.3-70B	Review agent checks plan, Evaluation agent scores it
Memory	JSON file	Saves user preferences across sessions
PDF Report	ReportLab	Generates the downloadable trip document